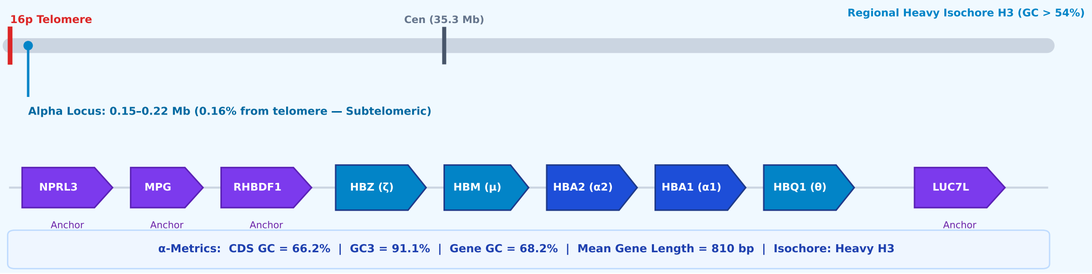
Figure 0: Conceptual Paradigm, Syntenic Discovery Pipeline, Metric Anatomy & Interactive Dashboard User Guide

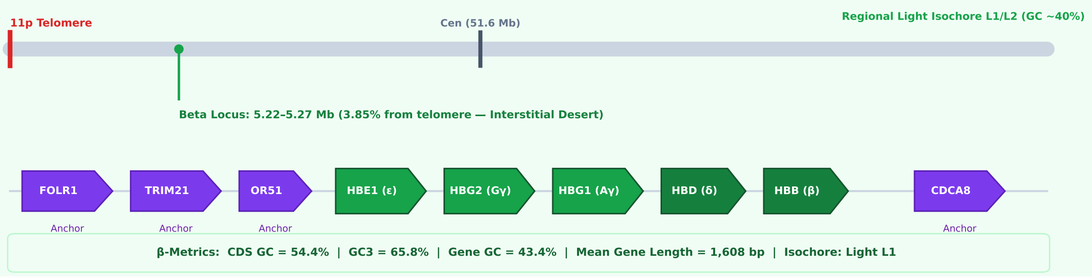
A. The Human Globin Paradigm & The Core Evolutionary Inquiry

Canonical dual-cluster arrangement on chromosomes 16 and 11, and the hypothesis of universal isochore polarity

Human Alpha-Globin Cluster — Subtelomeric Heavy Isochore (16p13.3)



Human Beta-Globin Cluster — Interstitial Light Isochore (11p15.4)



The Core Evolutionary Inquiry:

Is this dual-cluster, isochore-polarized architecture an ancient vertebrate feature, or a derived mammalian specialization?

How did 500 million years of chromosomal rearrangements, whole-genome duplications (2R/3R), and meiotic recombination shape globin synteny?

B. Synteny-Anchored Discovery & Karyotypic Tracking (68 Vertebrates)

Systematic discovery pipeline across 10 vertebrate clades spanning 500 million years of divergence

Step 1: Automated Syntenic Anchor Screening & Orthology Mapping

68 Genomes Screened across Tree of Life:

Agnatha • Chondrichthyes • Actinopterygii • Sarcopterygii • Amphibia • Testudines • Crocodylia • Aves • Squamata • Mammalia

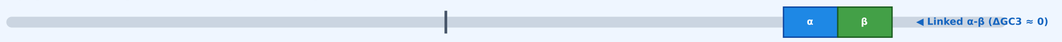
Target Syntenic Markers: α: NPRL3, MPG, RHBDF1, LUC7L, POLR3K | β: OR51, OR52, CDCA8, TRIM21, FOLR1, RRM1, STIM1

1. Primordial Linked Single Locus (Gar, Teleosts, Amphibians)

α and β globin genes physically paired in head-to-head tandem arrays on a single chromosome arm.

Identical chromatin context ensures parallel compositional evolution: ΔGC3 ≈ 0% (α = β).

Cytogenetic State: Single Chromosome (e.g. Gar LG13 / Frog Chr 3)



2. Canonical Decoupled Subtelomeric-α (Eutherian & Metatherian Mammals)

Amniote translocation decouples α and β. α captured by subtelomere (<1 Mb to telomere); β remains interstitial.

Runaway subtelomeric gBGC supercharges α-GC3 (>85–95%) while β-GC3 remains moderate (~60–65%). ΔGC3 >> 0%.

Cytogenetic State: Two Chromosomes (α Subtelomeric / β Interstitial)

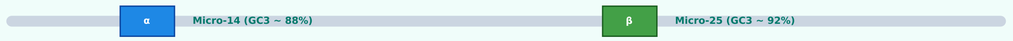


3. Dual Microchromosomal Partitioning (Archosauria & Aves — Chicken, Eagle)

Both α and β clusters translocate onto distinct, highly recombinogenic microchromosomes (Chr 14 and Chr 25).

High crossover density drives synchronous hyper-elevation of both clusters: α-GC3 > 85%, β-GC3 > 85%. ΔGC3 ≈ 0%.

Cytogenetic State: Two Microchromosomes (Micro-14 and Micro-25)



4. The Great Inversion (Squamate Reptiles — Lizards & Snakes)

Karyotypic inversion translocates β-locus to microchromosome/subtelomere, leaving α in an interstitial desert.

Inverted gBGC polarity: β-GC3 skyrockets (>90–96%) while α-GC3 languishes (~48–55%). ΔGC3 << 0% (β >> α)!

Cytogenetic State: Inverted Polarity (β Subtelomeric / α Interstitial)



C. Anatomy of Calculated Metrics: Gene Architecture & Codon Physics

Biophysical basis of CDS GC%, GC3%, Whole-Gene GC%, Intron Expansion, and Telomeric Gradients

1. Gene Architecture: Full Gene GC% vs. CDS GC% vs. Codon Positions

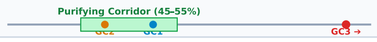


Whole-Gene GC%: Exons + Introns + UTRs (Reflects regional chromatin & isochore baseline GC)

CDS GC%: Coding sequence only (Concatenated Exons 1, 2, 3; reflects amino acid & mutational bias)

Codon Triplet Decomposition & Biophysical Roles:

- Position 1 (GC1): Moderate purifying constraint; conservative amino acid substitutions (~50–65% GC)
- Position 2 (GC2): Extreme purifying constraint! Strictly locked in Purifying Selection Corridor (45–55%)
- Position 3 (GC3): 70% synonymous/neutral! Pure biophysical barometer of recombination & gBGC (35–98% GC)
- 4-Fold Degenerate (GC4d): 100% synonymous (Gly, Ala, Val, Pro, Thr, Arg, Ser, Leu). Pure mutational/gBGC ground truth



2. Isochore Compaction Theory & Distance-to-Telomere Drivers

Bernardi's Genome Compaction Law:

High-GC isochores suppress intron expansion; low-GC isochores accumulate massive introns.

Human HBA1 (α-globin):



843 bp total (Intron 2 = 140 bp) [COMPACT HIGH-GC]

Human HBB (β-globin):



1,608 bp total (Intron 2 = 850 bp) [EXPANDED LOW-GC]

Distance-to-Telomere (dtelo / Lchr) as the Primary Engine of GC Polarity:

- Meiotic crossovers are physically biased toward chromosome ends (especially in male meiosis).
- Recombination-associated double-strand break repair preferentially resolves with GC alleles (gBGC).
- **Subtelomeric loci experience runaway GC3 elevation; interstitial loci remain buffered in AT-rich deserts.**

Relative Telomeric Distance Formula:

```
dtelo_rel = min(Locus_Pos, Chr_Len - Locus_Pos) / Chr_Len    (Range: 0.000 Subtelomeric  to  0.500 Metacentric/Internal)
```

D. Interactive Dashboard Architecture & Analytical User Guide

How to navigate and interpret the 5 interconnected exploratory modules in dashboard.html

Tab 1: Global Macroevolutionary Figures

The Big-Picture Evolutionary Synthesis

- Fig 1A: Curated benchmark summary across 33 representative species.
- Fig 1B: Exhaustive distribution across all 68 vertebrate genomes.
- Fig 2: Mechanistic drivers (Bivariate GC regressions, gene length compaction, telomere distance).
- Fig 3: Time-calibrated phylogenetic chronogram, chromosome synteny maps, and codon ΔGC3 divergence.

`UI Elements: [ Fig 1A ] [ Fig 1B ] [ Fig 2 ] [ Fig 3 ]`

Tab 2: 1-on-1 Head-to-Head Comparison

Pairwise Species Contrast & Polarity Scorecard

- Select Species A and Species B from full 68-genome searchable dropdowns.
- Live Comparative Scorecard: Instant ΔCDS GC, ΔGC3, and Isochore Polarity classification.
- Synchronized views: Linear Chromosome Tracks, Bivariate with Mini Inset, 6-Panel Diagnostic Distributions.
- Click any image or press Space/Enter to launch full-screen interactive pan & zoom modal.

`UI Elements: [ Species A: Human ▾ ] vs [ Species B: Corn Snake ▾ ] [ Inversion: ΔGC3 = -38% ]`

Tab 3: Single Species Deep Dive

High-Resolution Multi-Track Locus Inspection

- Inspect individual chromosome coordinates, centromere locations, and telomeric distances.
- View 6-panel diagnostic distributions contrasting globin loci against the whole-genome baseline.
- Focus on individual subpanels (Panels A–F) using keyboard shortcuts 1–6 in lightbox mode.

`UI Elements: [ 6-Panel Diagnostic Suite ] [ Interactive Lightbox Pan & Zoom ]`

Tabs 4 & 5: Interval Database & Statistical Rigor

Raw Gene Data & Hypothesis Testing

- Tab 4 (Interval Genes): Searchable table of all 1,180 globin and flanking syntenic genes with GC metrics.
- Tab 5 (Statistical Contrasts): Non-parametric Wilcoxon tests, Z-score distributions, and empirical percentiles.
- Downloadable CSV exports for independent bioinformatics replication.

`UI Elements: [ 1,180 Genes DB ] [ Non-Parametric Wilcoxon Tests ] [ Z-Scores ]`